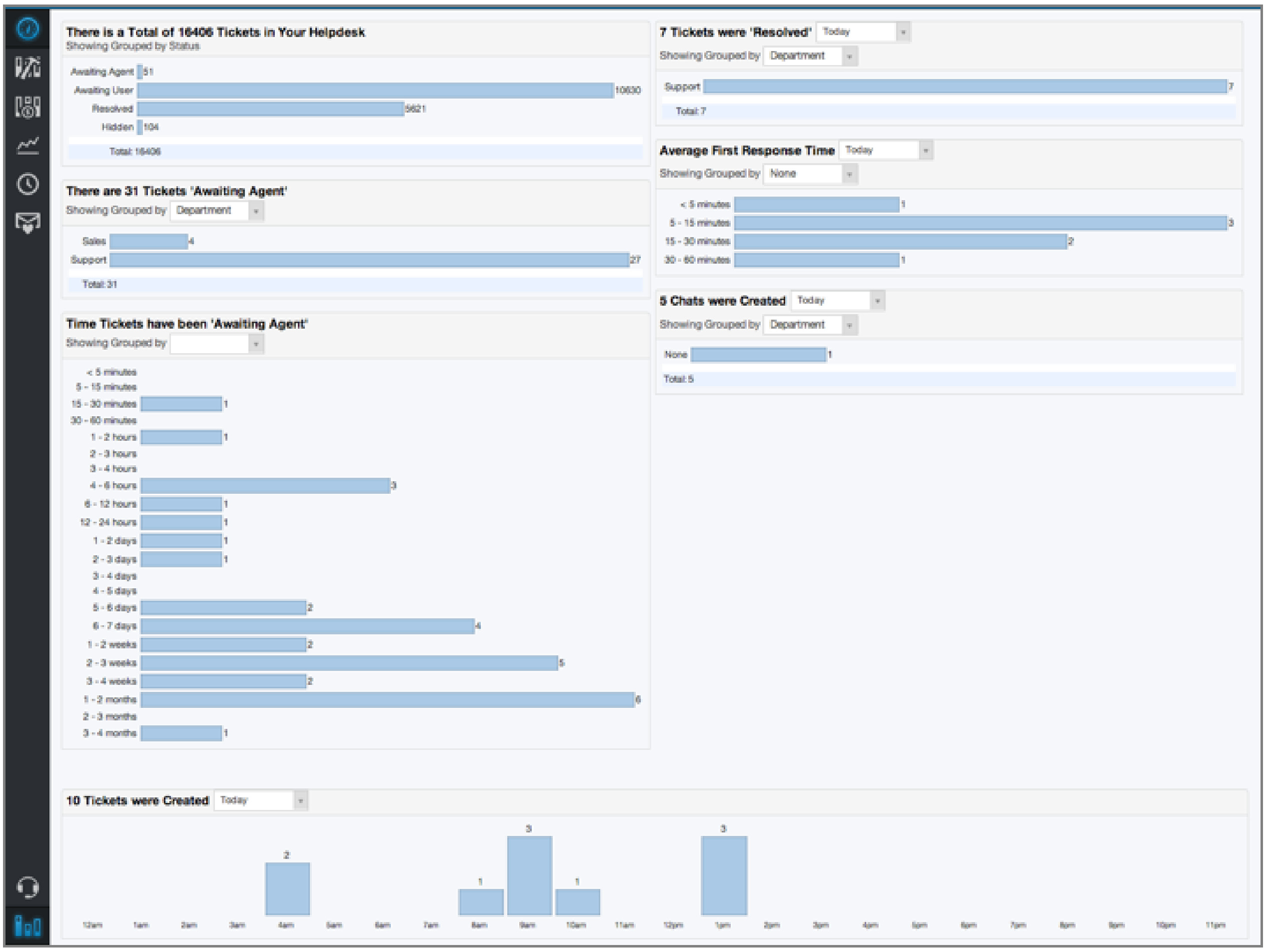
DeskPRO Reports Manual

July 2nd, 2014 • Revision #14

1. DeskPRO Reports Interface

DeskPRO includes a powerful report system.



This manual explains how to use the report interface, focusing on the report builder which allows you to generate detailed custom reports.

1.1. Using the report builder

Any agent who can access the reports interface can access the report builder through this icon:

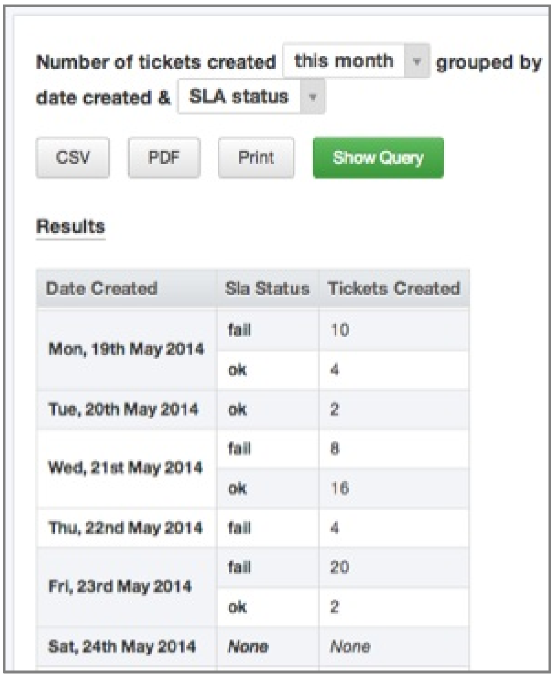
There are two types of reports in the report builder:

- **built-in**: specified by DeskPRO and cannot be edited directly
- **custom**: custom reports are specified by you or your agents

The data included in a report is specified with DPQL, the DeskPRO Query Language. See the DPQL Reference section for more details.

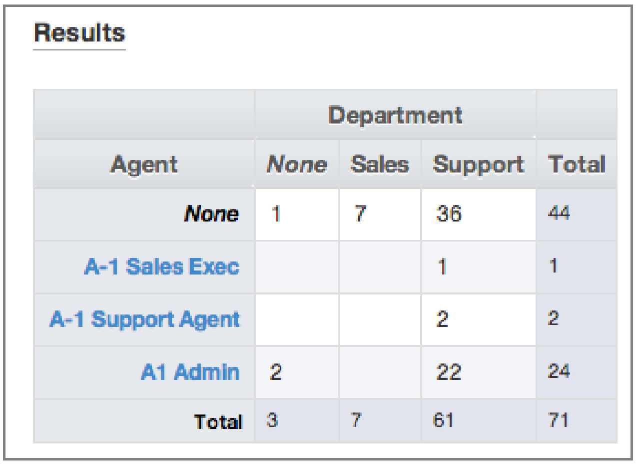
Reports can output information in several formats:

Tables - the most common output format is a table. In most cases, a simple table is generated, where a row represents one set of related data, such as the requested metrics for a particular department or agent.



Date Created	Sla Status	Tickets Created
Mon, 19th May 2014	fail	10
	ok	4
Tue, 20th May 2014	ok	2
Wed, 21st May 2014	fail	8
	ok	16
Thu, 22nd May 2014	fail	4
Fri, 23rd May 2014	fail	20
	ok	2
Sat, 24th May 2014	None	None

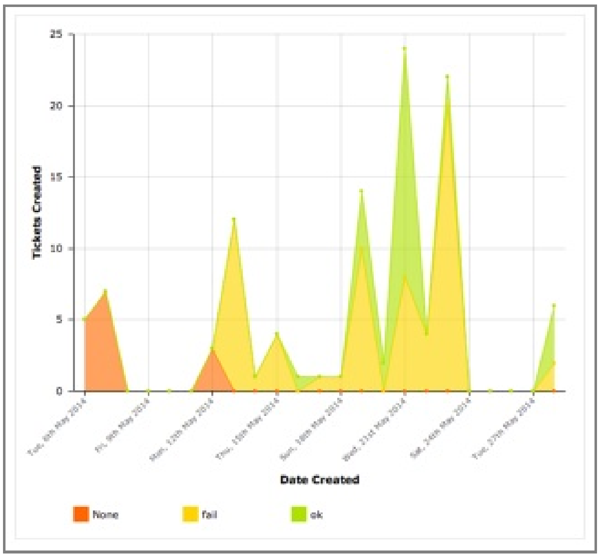
Queries can also generate "matrix tables", where specific data is listed across the top and left side of the table (such as departments and agents) and the values in the table represent a particular metric for that combination (such as a specific agent in a specific department).



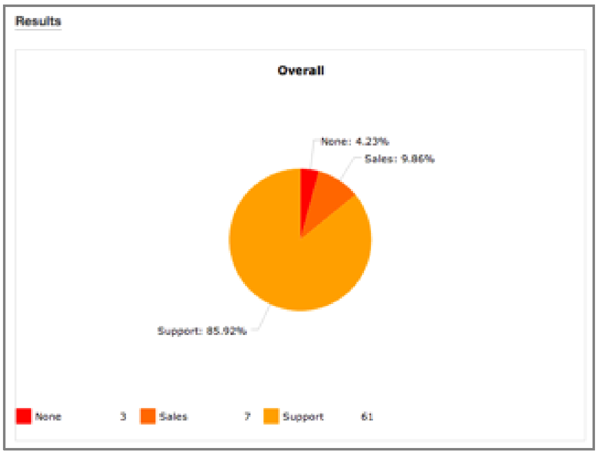
Results

	Department			
Agent	*None*	Sales	Support	Total
None	1	7	36	44
A-1 Sales Exec			1	1
A-1 Support Agent			2	2
A1 Admin	2		22	24
Total	3	7	61	71

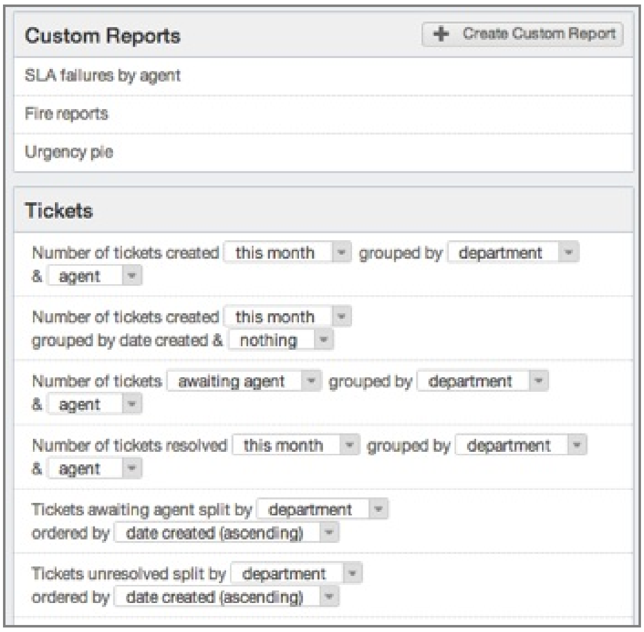
Bar and line chart - some reports display charts, showing a visual relationship for the requested metrics across the possible values.



Pie chart - the final type of chart is a pie chart that can be used to visualize relationships as a percentage. All values for a metric will be displayed as slices of the pie with their size depending on their portion of the whole.



To view a report, simply click on the name of it in the left reports interface pane. It will be run immediately and the results will be displayed in the main pane.



Similar reports will automatically be grouped together. To choose the particular report you wish to view in a group, click the underlined portion and choose the type you want to run. Additionally, built-in reports are grouped into categories.

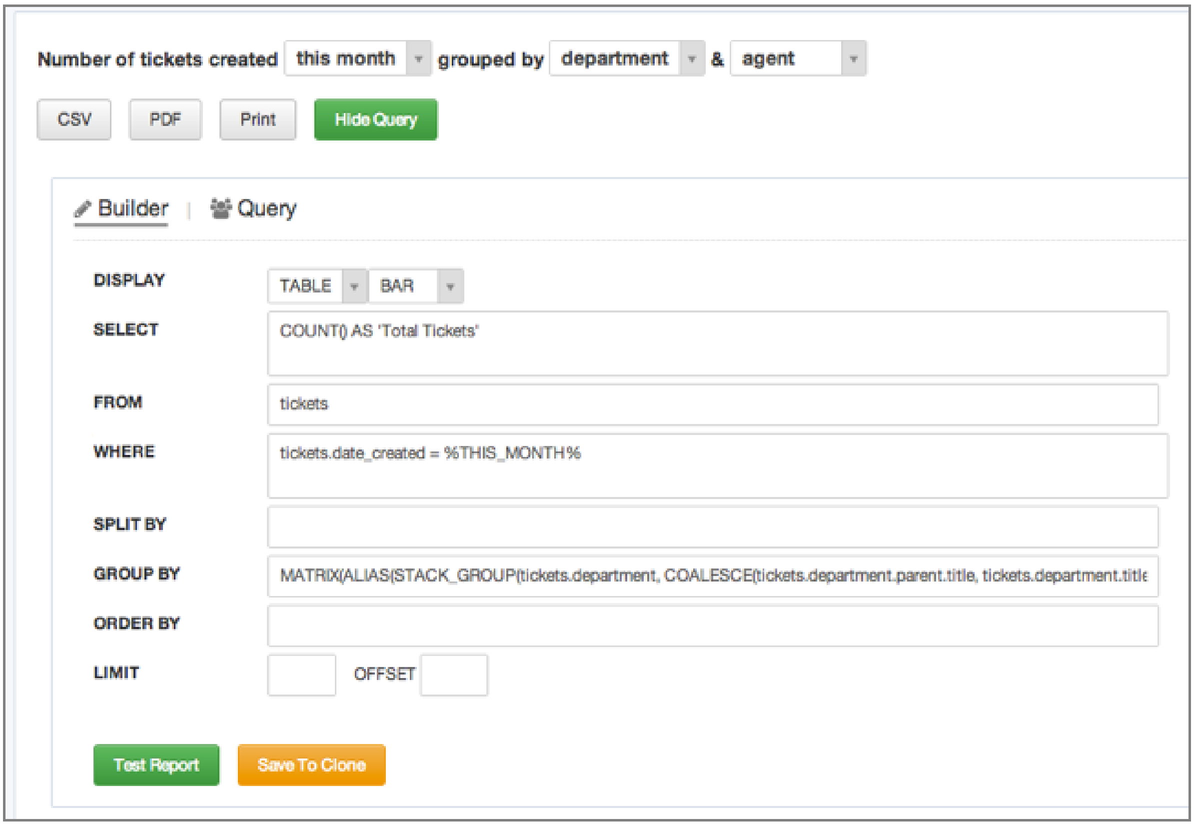
1.2. Report Actions

When you are viewing a particular report, you can take a number of actions:

- **CSV** - clicking this button will output the data in the report to a Comma Separated Values file which can be imported into spreadsheet software like Excel. Please note that CSV files can only contain text, so charts will not be included.
- **Print** - sends the report to your printer with simplified formatting
- **PDF** - saves the report as a PDF. The
- **Show Query** - displays the DPQL code which defines the report

You can use the **Show Query** view to edit the DPQL code for a report. Click **Test Report** to see the results of your changes.

When you're viewing a built-in report, you can't modify the report itself, but you can use **Save To Clone** to save the DPQL to a custom report.



2. DPQL Reference

The MySQL database which stores all your helpdesk data is organised into a series of tables.

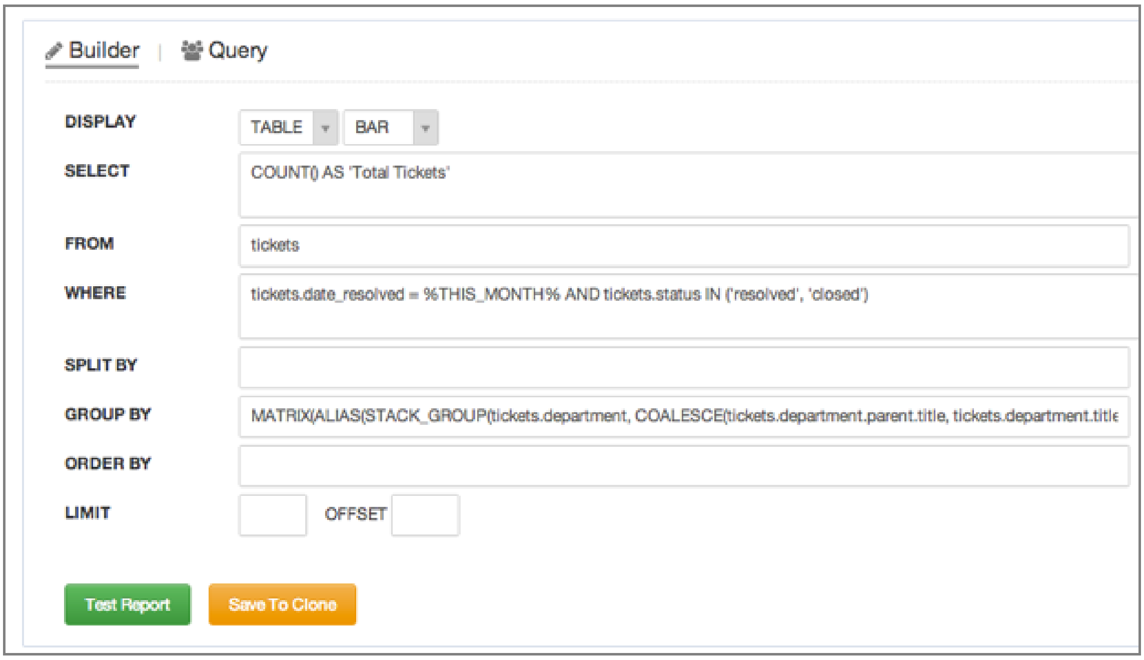
You can specify queries to retrieve data into reports using the DeskPRO Query Language (DPQL). DPQL has a syntax which resembles a simplified form of SQL commonly used to query databases.

Even if you don't know SQL, DPQL is relatively simple to use.

You can see the DPQL for the built-in reports. A good way to approach making a custom report is to clone a built-in report which does something similar, then modify it until it does what you want.

2.1. Anatomy of a DPQL query

Select a query and click the green **Show Query button** to see its DPQL. The **Builder** tab gives you a guided view; the **Query** tab lets you see the DPQL in plain text form.



DPQL queries consists of a series of clauses:

DISPLAY *display types*
SELECT *select expression*
FROM *database table*
[WHERE *conditions*]
[SPLIT BY *split fields*]
[GROUP BY *group fields*]
[ORDER BY *order fields*]
[LIMIT *amount* [OFFSET *offset amount*]]

- Clauses in [square brackets] are optional.
- *Italic text* is a placeholder for a value that you enter.

DISPLAY

The DISPLAY clause specifies how the output of the query is displayed, using the formats described in [Using the report builder](#).

It can take either one or two of the following values (separated with a comma if there are two):

- TABLE - displays the results as a table
- BAR - displays the results as a bar chart
- LINE - displays the results as a line graph
- PIE - displays the results as a pie chart

For example, `DISPLAY TABLE` means that the results specified by the rest of the query will be output in table form.

`DISPLAY BAR, TABLE` would output the results as a bar chart and a table.

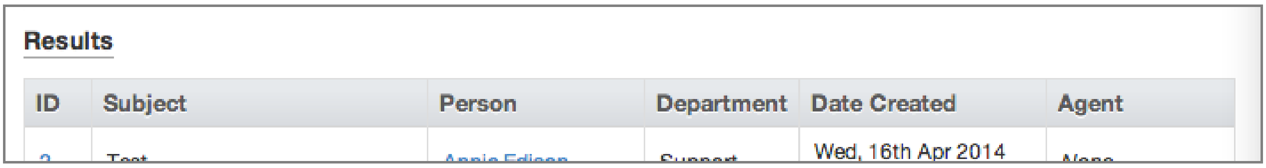
If you want to display a matrix table, you need to use the MATRIX() function in the GROUP BY clause.

SELECT

The SELECT clause works with the WHERE clause to define what information you want your report to include. It often contains **column references** which specify the data you want to retrieve from the table you choose in the FROM clause.

See the Appendix for a full list of columns.

For example, suppose you want to produce a table like this of all the tickets that meet a certain condition, e.g. awaiting agent:



Your SELECT clause would be a comma-separated list of column references:

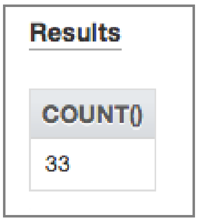
```
SELECT tickets.id, tickets.subject, tickets.person,
    tickets.department, tickets.date_created, tickets.agent
```

The SELECT clause can also use DPQL functions.

For example, suppose you didn't want a detailed table of all matching tickets, just the total number. You would use the COUNT() function to count the total number of matching tickets.

```
SELECT COUNT()
```

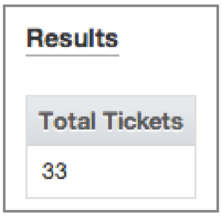
In table output, the header text for a column is automatically derived from the SELECT used to produce the column. This may not always result in a good column name; for example, if you just used `SELECT COUNT()` , you'll produce a table like this:



You can use an **alias** to specify a better name for a column.

```
COUNT() AS 'Total Tickets'
```

would make the result display like this:



FROM

The MySQL database which stores all your helpdesk data is organised into a series of tables. For example, the `tickets` table stores information about tickets.

The FROM clause specifies which database table your DPQL query is asking about. You can only pick one table.

All column references in the query must start with the table name specified in the FROM clause.

For a list of valid tables, see the Tables section in the appendix.

WHERE

The conditions in the optional WHERE clause are used to limit what data is displayed or used in a calculation.

For example, if your query was

```
DISPLAY TABLE
FROM tickets
SELECT tickets.id
```

you'd get a table with every ticket's ID - not very useful.

But with:

```
DISPLAY TABLE
FROM tickets
SELECT tickets.id
WHERE tickets.status = 'awaiting_agent'
```

you'd get a table with the IDs of just the tickets with a status of Awaiting Agent.

Conditions can be joined by operators such as AND, OR, NOT, IN and parentheses (brackets) to make complex expressions.

For example:

```
WHERE tickets.status IN ('awaiting_agent','awaiting_user')
```

will return all tickets with either Awaiting Agent or Awaiting User status.

```
tickets.labels.label != NULL
```

will return all tickets that have a label (i.e. the label value does not equal NULL).

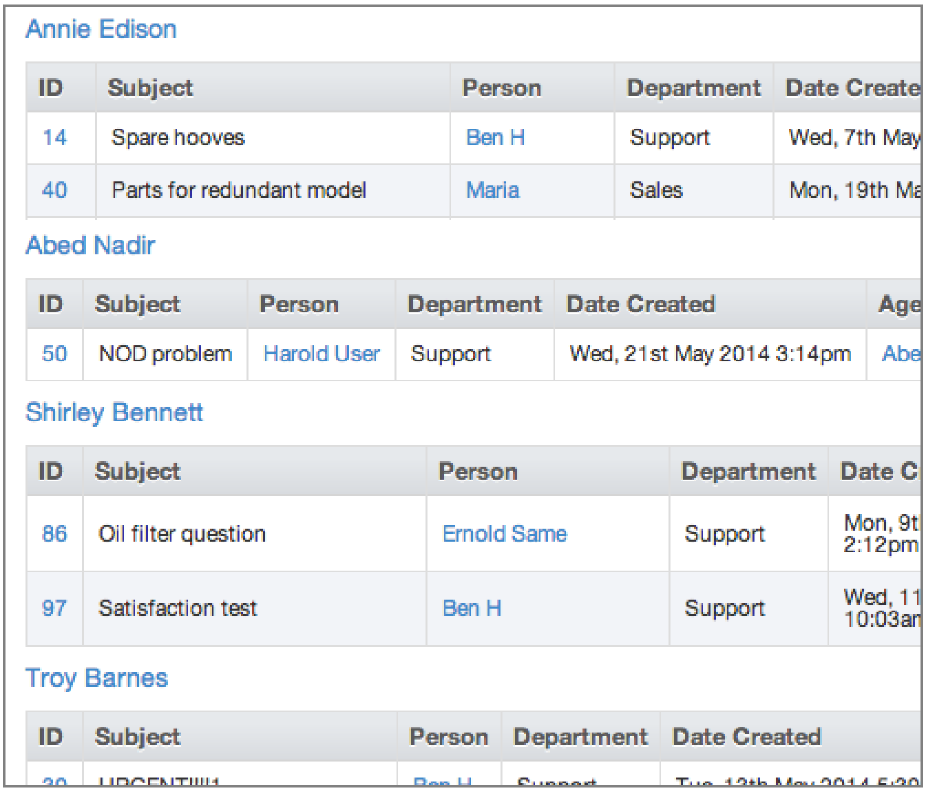
SPLIT BY

The optional SPLIT BY clause enables you split the results into separate tables/graphs by providing split field values.

For example, if you wanted to display each agent's matching tickets for a query in a separate table, you'd use:

```
SPLIT BY tickets.agent
```

and the result would look like this:



Annie Edison

ID	Subject	Person	Department	Date Create
14	Spare hooves	Ben H	Support	Wed, 7th May
40	Parts for redundant model	Maria	Sales	Mon, 19th Ma

Abed Nadir

ID	Subject	Person	Department	Date Created	Age
50	NOD problem	Harold User	Support	Wed, 21st May 2014 3:14pm	Abe

Shirley Bennett

ID	Subject	Person	Department	Date C
86	Oil filter question	Ernold Same	Support	Mon, 9t 2:12pm
97	Satisfaction test	Ben H	Support	Wed, 11 10:03ar

Troy Barnes

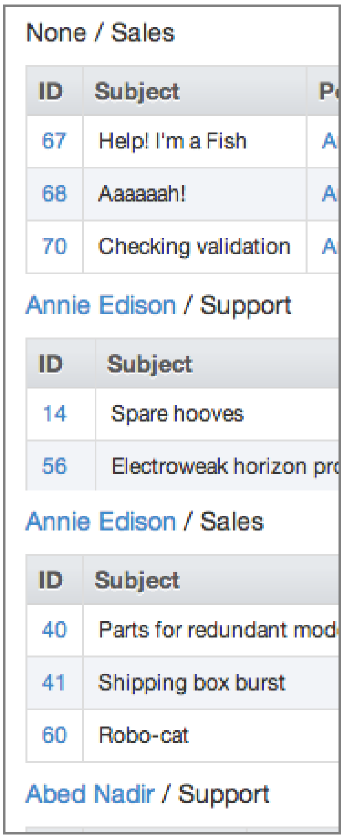
ID	Subject	Person	Department	Date Created

with a separate table for each agent.

You can use multiple split fields by providing a comma-separated list of expressions. This will result in separate tables for each combination of of the fields you provide. For example:

```
SPLIT BY tickets.agent, tickets.department
```

produces output like this:



None / Sales

ID	Subject	P
67	Help! I'm a Fish	A
68	Aaaaaah!	A
70	Checking validation	A

Annie Edison / Support

ID	Subject
14	Spare hooves
56	Electroweak horizon pr

Annie Edison / Sales

ID	Subject
40	Parts for redundant mod
41	Shipping box burst
60	Robo-cat

Abed Nadir / Support

with separate tables for Agent A/Department A, Agent A/Department B, Agent B/Department A, etc.

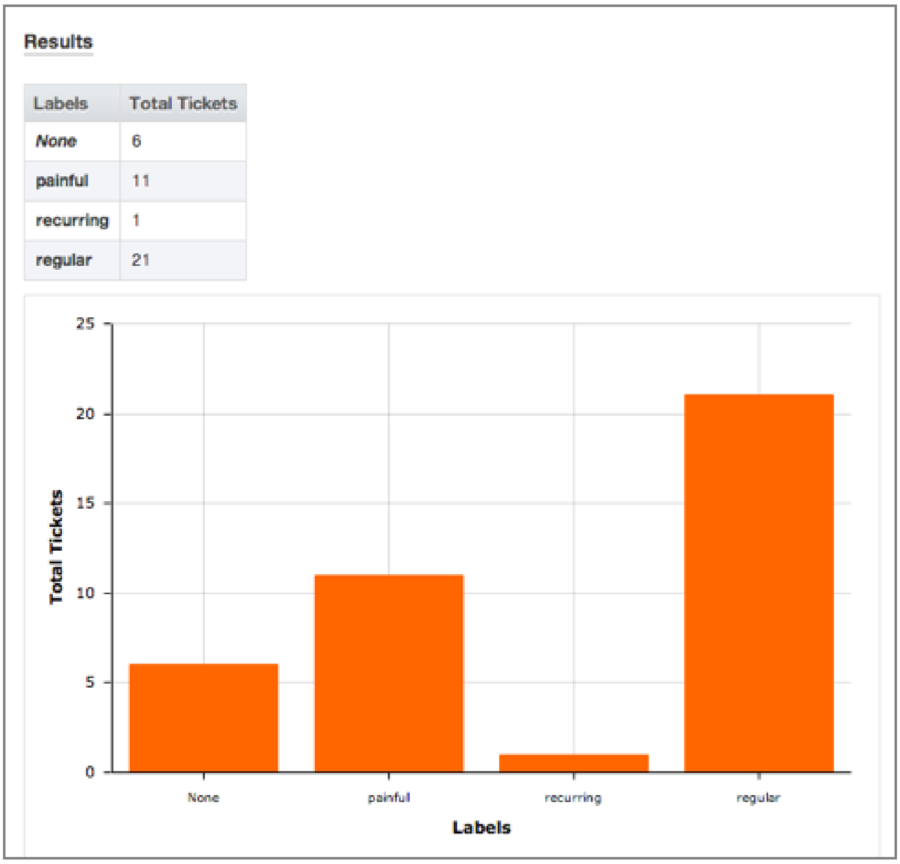
GROUP BY

The GROUP BY clause is used to group records that have the same values for the specified fields.

For example, you can group by ticket labels:

```
DISPLAY TABLE, BAR
SELECT COUNT() AS 'Total Tickets'
FROM tickets
WHERE tickets.status = 'awaiting_agent'
GROUP BY tickets.labels
```

In the output, tickets with the same label are grouped into table rows or bars in the chart:



You can also use groups to do calculations such as determining totals and averages.

The group by fields will automatically be displayed in the resulting table, so there is no need to add them to the SELECT clause.

As in the SELECT clause, group fields can be aliased with `AS 'Label'` to change the name of the table header row.

To create a matrix table, use the MATRIX() function in the GROUP BY clause to specify two groups. The first specifies the values going across the top of the table, while the second controls the values going down the left side.

ORDER BY

The ORDER BY clause determines how the rows will be ordered when they are returned. If no order is given, the results will be displayed in an undefined order.

The order fields are comma-separated expressions. Ordering will happen across the fields from left to right (ordering by the first expression, then using the second to resolve ties, and so on). Each expression may optionally have `ASC` or `DESC` appended to it to control whether ordering is in ascending or descending order. (Ascending is the default if no direction is specified).

For example:

```
ORDER BY SUM(tickets.total_user_waiting) DESC
```

would order a list of users or organizations by their total waiting time, with the highest waiting time at the top.

The ORDER BY clause can access aliases that were specified in the SELECT or GROUP BY clauses using the syntax `@'alias'` (for example: `@'Total Tickets'`). Referencing an alias causes the results to be ordered as if you had written the aliased expression in the ORDER BY clause.

For example, suppose you want to make a table showing how many tickets each agent has, sorted in descending order:

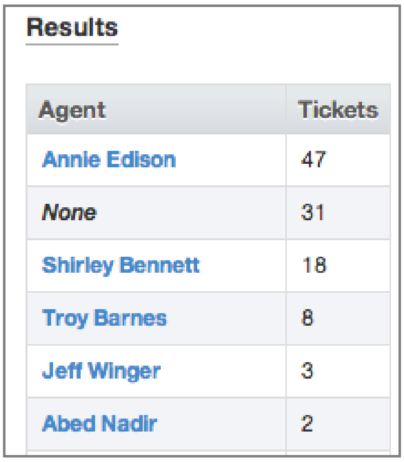
```
DISPLAY TABLE

SELECT COUNT() AS 'Tickets'

FROM tickets

GROUP BY tickets.agent

ORDER BY @'Tickets' DESC
```


Results

Agent	Tickets
Annie Edison	47
None	31
Shirley Bennett	18
Troy Barnes	8
Jeff Winger	3
Abed Nadir	2

LIMIT / OFFSET

The LIMIT clause enables you to limit the number of rows returned. If you don't specify a limit amount, a default limit of 2500 will be used to ensure correct operation.

You can use the OFFSET clause to skip over a certain number of rows before returning the LIMIT *amount* of rows. The OFFSET defaults to 0.

2.2. General expression format

Most clauses of a DPQL statement accept a general expression format. You can use expressions to carry out more complicated queries.

Complex expressions are made up of smaller, simpler expressions. Expressions are made up of the following components:

Type	Example	Details
Numbers	5, 37.4	Simple, literal references to integer or decimal values
Strings	'string' or "string"	Strings are literal references to text. These will commonly be used in tests/comparisons and aliases.
Null	tickets.labels.label != NULL	You can use this in tests to check whether a certain value exists. If a certain value has been set, it will be not equal to NULL.
Parentheses	(tickets + chat_conversations)/agents	Parentheses are used to enforce the order in which the expression is evaluated.
Column references	table.col[.col2...]	Column references directly retrieve data from your helpdesk. tickets.subject and tickets.agent.name are both valid column references. You can use multiple column parts to combine data from different tables based on a shared field between them (the equivalent of an SQL join). For example, tickets.person.organization.name which would retrieve the name of the organization the person that started the ticket belongs to.
Function calls	COUNT(), CURDATE(), MATRIX(group X, group y)	These are useful functions, for example doing calculations on the inputs (arguments) you give them, changing formatting, or retrieving information. COUNT() is a function. Some functions don't need any input, e.g. CURDATE() just gets the current date.
Comparisons	(expression) [=, !=, >=, >, <, <=] (expression)	Enable you to compare two expressions e.g. tickets.count_agent_replies >= 10 checks if the number of agent replies on a ticket is greater than or equal to 10. If the comparison is true, the value of the comparison = 1. If it's false, it = 0.
Placeholders	%TODAY%, %LAST_YEAR%	Placeholders are dynamic elements of a DPQL query that are automatically updated to the appropriate value when the query is run. They are useful in comparisons e.g. you can use tickets.created_date = %PAST_7_DAYS% would match tickets created over the last 7 days.
Logic	AND, OR	Used to logically combine two expressions.

2.3. Date references and comparisons

You can reference specific dates directly in your queries, rather than using the relative date placeholders. Dates can be referenced in two formats:

YYYY-MM-DD - e.g. `2013-10-31` ; refers to a date only. This implicitly has a time of 00:00:00 of the specified day.

YYYY-MM-DD HH:MM:SS - e.g. `2012-10-31 23:35:52` ; refers to a date and a specific time in a 24-hour format.

Dates and times used in queries automatically reflect the timezone of the person running them.

This is an example DPQL query to list the tickets created from October 1st to 15th, 2012:

```
DISPLAY TABLE
SELECT tickets.id
FROM tickets
WHERE tickets.date_created >= '2012-10-01'
AND tickets.date_created <= '2012-10-15 23:59:59'
```

2.4. List of placeholders

%EVER%	A range covering any date.
%LAST_MONTH%	A range covering the entirety of the previous month. For example, if today is any day in September 2012, this placeholder will match all of August 2012.
%LAST_WEEK%	A range covering the entirety of the previous week, based on a new week starting on a Monday. For example, if today is Wednesday the 12th, this will match Monday the 3rd to Sunday the 9th.
%LAST_YEAR%	A range covering the entirety of the previous year. For example, if today is any day in 2012, this placeholder will match all of 2011.
%PAST_24_HOURS%	A range covering from exactly 24 hours ago until when the report is run.
%PAST_7_DAYS%	A range covering the 7 full days prior to the day when the report is run.
%PAST_30_DAYS%	A range covering the 30 full days prior to the day when the report is run.
%THIS_MONTH%	A range covering the entirety of the current month, from 00:00 on the first day until 23:59 on the last day.
%THIS_WEEK%	A range covering the entirety of the current week, from 00:00 on the first day until 23:59 on the last day. Weeks are assumed to start on a Monday and end on a Sunday
%THIS_YEAR%	A range covering the entirety of the current year, from 00:00 on the first day until 23:59 on the last day.
%TODAY%	A range covering the entirety of the today.
%TOMORROW%	A range covering the entirety of the tomorrow.
%YESTERDAY%	A range covering the entirety of the yesterday.

2.5. List of functions

COUNT([condition])	Counts the total number of rows matched by the query or current group. If a condition is provided, counts the number of rows in the query or current group that match the condition.
COUNT_DISTINCT(expression)	Counts the total number distinct values for the given expression within the rows matched by the query or current group.
CURDATE()	Gets the current date in the time zone of the person running the query.
CURTIME()	Gets the current time in the time zone of the person running the query.
DATE_OFFSET_GROUP(seconds)	Groups a date offset/difference in seconds into human-readable ranges (0-15 minutes, 15-30 minutes, etc).
DATE_OFFSET_GROUP(to date, from date)	Displays the difference between the two provided dates as human-readable ranges (0-15 minutes, 15-30 minutes, etc).
FORMAT(value, format)	formats the value into the specified format. Possible formats include boolean, number, numberraw, datetime, date, time, year, percent, string.
LINK(value, link[, params..])	Links to value using the URL provided in link. Placeholders in link are replaced by the additional params. Placeholder values should be represented in sprintf format.
MATRIX(group X, group Y)	If both provided groups are non-null creates a matrix table from them. Otherwise, it creates a standard grouped table.
NOW()	Gets the current date and time in the time zone of the person running the query.
PERCENT(condition[, decimals])	determines the percentage of total rows or rows within the current group that match the condition The results are displayed as a percentage with decimals controlling the precision. 2 decimals are shown by default.
PERCENT(sql, printed)	gets the sql but ensures that printed is used if the value is ever going to be displayed. This is mostly helpful in the GROUP BY clause where you need to group on one expression but display the results of another.
TO_UTC(date)	converts date to UTC from the current person's time zone.
UTC(expression)	ensures that all dates and times within this function are calculated using UTC. This can increase performance.

A large number of functions are also available that have the exact same behavior as their MySQL equivalents:

- ABS
- AVG
- CEILING

- CONCAT
- CONCAT_WS
- CHAR_LENGTH
- DATE
- DATE_FORMAT
- DATEDIFF
- DAYNAME
- DAYOFMONTH
- DAYOFWEEK
- DAYOFYEAR
- FIELD
- FIND_IN_SET
- FLOOR
- FROM_UNIXTIME
- GREATEST
- HOUR
- IF
- IFNULL
- ISNULL
- LAST_DAY
- LEAST
- LEFT
- LENGTH
- LOCATE
- LOWER
- LPAD
- LTRIM
- MAX
- MIN
- MINUTE
- MONTH
- MONTHNAME
- POW
- QUARTER
- RAND
- REPEAT

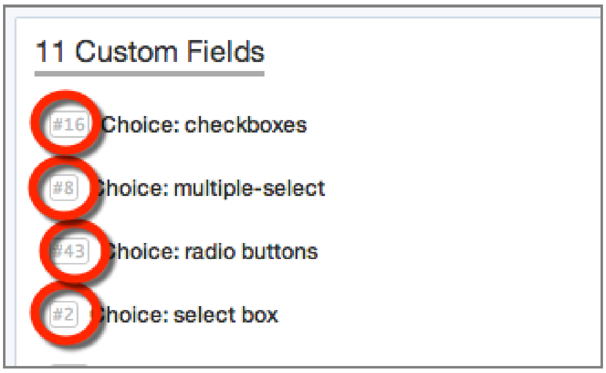
- REPLACE
- REVERSE
- RIGHT
- ROUND
- RPAD
- RTRIM
- SECOND
- SQRT
- STDDEV_POP
- STDDEV_SAMP
- STRCMP
- SUBSTRING
- SUBSTRING_INDEX
- SUM
- TIME
- TIMESTAMP
- TRIM
- TRUNCATE
- UNIX_TIMESTAMP
- UPPER
- UTC_DATE
- UTC_TIME
- UTC_TIMESTAMP
- VAR_POP
- VAR_SAMP
- WEEKDAY
- WEEKOFYEAR
- YEAR

2.6. Accessing custom fields

Several content types in DeskPRO, including tickets and articles, support custom fields. You can access these fields in custom reports using the following syntax.

Each custom field has an ID number. In the admin interface, click **View Options** in the top left of the toolbar and enable **Show IDs next to items**

You will then see the ID number in the relevant part of the admin interface. For example, for custom ticket fields, look in **Tickets** > **Fields**. Make note of the ID as it will be needed in your query.



Once you have the ID, you can reference the value of a particular field via `custom_data[#]`. For example, to refer to a custom ticket field with ID number 12, you would use `tickets.custom_data[12]`.

This query will give a count of all tickets created this month per value for the custom field with ID 12:

```
DISPLAY TABLE
SELECT COUNT() AS 'Total Tickets'
FROM tickets
WHERE tickets.date_created = %THIS_MONTH%
GROUP BY tickets.custom_data[12]
```

This will list the tickets created this month, split by the values of the custom field with ID 12:

```
DISPLAY TABLE, BAR
SELECT tickets.id, tickets.subject, tickets.person
FROM tickets
WHERE tickets.date_created = %THIS_MONTH%
SPLIT BY tickets.custom_data[12]
```

2.7. Advanced: Dynamic replacement

When creating custom reports, you can set up specific values to be dynamically replaced by a user selection. This is used by the built-in reports to allow users to choose a date range, grouping field, ticket status, or order.

Each dynamic replacement has two components:

- The title - this is how the user selects which value they want for the dynamic replacement. A default value should be specified here. The specific format varies for each replacement type, but the general format is `<#:replacement type[additional options]>` where # is a unique ID to identify the corresponding replacement in the query.
- The query - this is where the actual replacement is made to run the correct query. The general form is `%#:REPLACEMENT_TYPE[additional options]%` where # corresponds to the correct replacement in the title.

The available replacements are as follows:

Date Ranges

Title format	<#:date group, default: *default value*>
Possible defaults	today, yesterday, this_week, this-month, this_year, last_week, last_month, last_year, past_24_hours, past_7_days, past_30_days, ever
Partial title example	<1:date group, default: *this_month*>
Query format	%#:DATE_GROUP%
Partial query example	tickets.date_created = %1:DATE_GROUP%

Ticket Statuses

Title format	<#:status group:tickets, default: *default value*>
Possible defaults	awaiting_user, awaiting_agent, unresolved, resolved, hidden, any
Partial title example	<1:status group:tickets, default: awaiting_agent>
Query format	%#:STATUS_GROUP:tickets:*table_name*% (where *table_name* is how you access the tickets table in the query)
Partial query example	%1:STATUS_GROUP:tickets:tickets_messages.ticket%

Field Groups

Title format	<#:field group:tickets, default: *default value*>

Possible defaults	department, agent, agent_team, person, organization, language, usergency, category, priority, workflow, sla, sla_status, hour_created, day_week_created, day_month_created, month_created, year_created, *ticketfield#* (for custom ticket fields), *personfield#* (for custom person fields), *orgfield#* (for custom organization fields), none
Partial title example	grouped by <2:field group:tickets, default: department> & <3:field group:tickets, default: agent>
Query format	%#:FIELD_GROUP:tickets:*table_name*% (where *table_name* is how you access the tickets table in the query)
Partial query example	MATRIX(%2:FIELD_GROUP:tickets:tickets%, %3:FIELD_GROUP:tickets:tickets%)

Ordering

Title format	<#:field group:tickets, default: *default value*>
Possible defaults	date_created_asc, date_created_desc, last_agent_reply_asc, last_agent_reply_desc, last_user_reply_asc, last_user_reply_desc, total_waiting_asc, total_waiting_desc
Partial title example	ordered by <2:order group:tickets, default: date_created_asc>
Query format	%#:ORDER_GROUP:*table_name*% (where *table_name* is how you access the tickets table in the query)
Partial query example	%2:ORDER_GROUP:tickets%

3. Example Reports

Here are some worked examples of making reports based on common requests from DeskPRO customers.

3.1. Reporting ticket backlog

Requirement: a report showing the number of unresolved tickets that are over 7, 14 and 30 days old.

```
DISPLAY TABLE
SELECT COUNT() AS 'Total',
    COUNT(tickets.date_created < %PAST_7_DAYS%) AS 'Unresolved over 7 Days',
    COUNT(UNIX_TIMESTAMP(tickets.date_created) < (UNIX_TIMESTAMP(NOW()) - (14 * 24 * 60 * 60))) AS 'U
    COUNT(tickets.date_created < %PAST_30_DAYS%) AS 'Unresolved over 30 Days'
FROM tickets
WHERE tickets.status IN ('awaiting_user', 'awaiting_agent')
```

It's fairly simple to use the built-in `%PAST_7_DAYS%` and `%PAST_30_DAYS%` placeholders to find tickets that were created over 7 days or 30 days ago, but there isn't a placeholder for 14 days.

We need a way to compare the date/time of tickets were created with the date/time of 14 days ago.

One way to implement this is to convert the `tickets.date_created` into Unix timestamp format using `UNIX_TIMESTAMP(tickets.date_created)` . Unix timestamp format counts the number of seconds since an arbitrary date/time.

We can then compare that value with the Unix timestamp for 14 days ago. We use the NOW() function to get the current date and time (not forgetting to convert that into a Unix timestamp too), then subtract the number of seconds in 14 days.

The `WHERE tickets.status IN ('awaiting_user', 'awaiting_agent')` clause limits the tickets to those with Awaiting User or Awaiting Agent status, i.e. those which are unresolved. You might think that we should include On Hold status in the list, but tickets on hold still have `tickets.status` of `awaiting_agent` - hold status is recorded with a separate `is.hold` value.

[more examples to come]

4. APPENDIX: DPQL Field Reference

This appendix contains a complete list of the available tables and columns, as well as the type of data that each column contains.

4.1. Tables

Table Name	Description
articles	Articles and information in the knowledgebase
article_attachments	Attachments to knowledgebase articles
article_comments	Comments for each knowledgebase article
chat_conversations	Records for each chat
chat_messages	Individual messages in each chat
downloads	Information about each file that has been specified as a download
download_comments	Individual comments for each download
feedback	Customer feedback and suggestion records
feedback_attachments	Attachments to feedback and suggestions
feedback_comments	Comments for feedback and suggestions
labels_articles	Labels for knowledgebase articles
labels_chat_conversations	Labels for chats
labels_downloads	Labels for downloads
labels_feedback	Labels for feedback and suggestions
labels_news	Labels for news entries
labels_organizations	Labels for organizations
labels_people	Labels for registered people
labels_tasks	Labels for tasks
labels_tickets	Labels for tickets
news	News entries
news_comments	Comments on news entries
organizations	Organizations
people	Registered people (users and agents)
people_emails	Email addresses in use by registered people
tasks	Tasks
task_comments	Comments on tasks
tickets	Tickets
tickets_log	Ticket change log entries

tickets_messages	Individual messages in tickets
ticket_attachments	Attachments to tickets
ticket_charges	Ticket billing charges
ticket_feedback	Feedback on ticket responses
ticket_slas	SLA status records for tickets

4.2. Columns and data types

Agent Team

Field Name	Data Type
id	number
name	string

Article

Field Name	Data Type
content	string
date_created	datetime
date_end	datetime
date_published	datetime
end_action	string
hidden_status	string
id	number
language_id	number
num_comments	number
num_ratings	number
person_id	number
slug	string
status	string
title	string
total_rating	number
view_count	number
attachments	Article Attachment
custom_data	Custom Data Article
custom_data[#]	Custom Data Article (Gets data for the article field with ID #)
labels	Label Article
language	Language

person	Person
revisions	Article Revision

Article Attachment

Field Name	Data Type
article_id	number
blob_id	number
date_created	datetime
id	number
person_id	number
article	Article
blob	Blob
person	Person

Article Comment

Field Name	Data Type
article_id	number
content	string
date_created	datetime
email	string
id	number
ip_address	string
is_reviewed	boolean
name	string
person_id	number
status	string
validating	string
visitor_id	number
website	string
article	Article

person	Person
visitor	Visitor

Article Revision

Field Name	Data Type
article_id	number
content	string
date_created	datetime
id	number
person_id	number
status	string
title	string
article	Article
person	Person

Blob

Field Name	Data Type
authcode	string
blob_hash	string
content_type	string
date_cleanup	datetime
date_created	datetime
dim_h	number
dim_w	number
filename	string
filesize	number
id	number
is_media_upload	boolean
is_temp	boolean
original_blob_id	number

save_path	string
storage_loc	string
sys_name	string
title	string
labels	Label Blob
original_blob	Blob

Chat Conversation

Field Name	Data Type
agent_id	number
agent_team_id	number
date_assigned	datetime
date_created	datetime
date_ended	datetime
date_first_agent_message	datetime
department_id	number
ended_by	string
id	number
is_agent	boolean
is_window	boolean
person_email	string
person_id	number
person_name	string
rating_comment	string
rating_overall	number
rating_response_time	number
session_id	number
status	string
subject	string
total_to_ended	number

visitor_id	number
agent	Person
agent_team	Agent Team
department	Department
labels	Label Chat Conversation
messages	Chat Message
person	Person
session	Session
visitor	Visitor

Chat Message

Field Name	Data Type
author_id	number
content	string
conversation_id	number
date_created	datetime
id	number
is_html	boolean
is_sys	boolean
is_user_hidden	boolean
metadata	string
origin	string
person_name	string
author	Person
conversation	Chat Conversation

Custom Data Article

Field Name	Data Type
article_id	number
field_id	number

id	number
input	string
root_field_id	number
value	number
article	Article
field	Custom Def Article
root_field	Custom Def Article

Custom Data Feedback

Field Name	Data Type
feedback_id	number
field_id	number
id	number
input	string
root_field_id	number
value	number
feedback	Feedback
field	Custom Def Feedback
root_field	Custom Def Feedback

Custom Data Organization

Field Name	Data Type
field_id	number
id	number
input	string
organization_id	number
root_field_id	number
value	number
field	Custom Def Organization
organization	Organization

root_field	Custom Def Organization

Custom Data Person

Field Name	Data Type
field_id	number
id	number
input	string
person_id	number
root_field_id	number
value	number
field	Custom Def Person
person	Person
root_field	Custom Def Person

Custom Data Ticket

Field Name	Data Type
field_id	number
id	number
input	string
root_field_id	number
ticket_id	number
value	number
field	Custom Def Ticket
root_field	Custom Def Ticket
ticket	Ticket

Custom Def Article

Field Name	Data Type
default_value	string
description	string

display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Article
parent	Custom Def Article
plugin	Plugin

Custom Def Feedback

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string

options	string
parent_id	number
plugin_id	number
sys_name	string
title	string
children	Custom Def Feedback
parent	Custom Def Feedback
plugin	Plugin

Custom Def Organization

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Organization
parent	Custom Def Organization
plugin	Plugin

Custom Def Person

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean
is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Person
parent	Custom Def Person
plugin	Plugin

Custom Def Ticket

Field Name	Data Type
default_value	string
description	string
display_order	number
handler_class	string
has_display_template	boolean
has_form_template	boolean
id	number
is_agent_field	boolean

is_enabled	boolean
is_user_enabled	boolean
js_class	string
options	string
parent_id	number
plugin_id	number
title	string
children	Custom Def Ticket
parent	Custom Def Ticket
plugin	Plugin

Department

Field Name	Data Type
display_order	number
email_gateway_id	number
id	number
is_chat_enabled	boolean
is_tickets_enabled	boolean
parent_id	number
title	string
user_title	string
children	Department
email_gateway	Email Gateway
parent	Department

Download

Field Name	Data Type
blob_id	number
category_id	number
content	string

date_created	datetime
date_published	datetime
hidden_status	string
id	number
language_id	number
num_comments	number
num_downloads	number
num_ratings	number
person_id	number
slug	string
status	string
title	string
total_rating	number
view_count	number
blob	Blob
category	Download Category
labels	Label Download
language	Language
person	Person
revisions	Download Revision

Download Category

Field Name	Data Type
depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	Download Category

parent	Download Category

Download Comment

Field Name	Data Type
content	string
date_created	datetime
download_id	number
email	string
id	number
ip_address	string
is_reviewed	boolean
name	string
person_id	number
status	string
validating	string
visitor_id	number
website	string
download	Download
person	Person
visitor	Visitor

Download Revision

Field Name	Data Type
blob_id	number
content	string
date_created	datetime
download_id	number
id	number
person_id	number
status	string

title	string
blob	Blob
download	Download
person	Person

Email Gateway

Field Name	Data Type
connection_options	string
connection_type	string
date_last_check	datetime
department_id	number
gateway_type	string
id	number
is_enabled	boolean
keep_read	boolean
linked_transport_id	number
title	string
addresses	Email Gateway Address
department	Department
linked_transport	Email Transport

Email Gateway Address

Field Name	Data Type
email_gateway_id	number
id	number
match_pattern	string
match_type	string
run_order	number
gateway	Email Gateway

Email Source

Field Name	Data Type
blob_id	number
date_created	datetime
error_code	string
gateway_id	number
headers	string
header_from	string
header_subject	string
header_to	string
id	number
object_id	number
object_type	string
source_info	string
status	string
uid	string
blob	Blob
gateway	Email Gateway

Email Transport

Field Name	Data Type
id	number
match_pattern	string
match_type	string
run_order	number
title	string
transport_options	string
transport_type	string

Feedback

Field Name	Data Type

category_id	number
content	string
date_created	datetime
date_published	datetime
hidden_status	string
id	number
language_id	number
num_comments	number
num_ratings	number
person_id	number
popularity	number
slug	string
status	string
status_category_id	number
title	string
total_rating	number
validating	string
view_count	number
attachments	Feedback Attachment
category	Feedback Category
comments	Feedback Comment
custom_data	Custom Data Feedback
custom_data[#]	Custom Data Feedback (Gets data for the feedback field with ID #)
labels	Label Feedback
language	Language
person	Person
revisions	Feedback Revision
status_category	Feedback Status Category

Feedback Attachment

Field Name	Data Type
blob_id	number
date_created	datetime
feedback_id	number
id	number
person_id	number
blob	Blob
feedback	Feedback
person	Person

Feedback Category

Field Name	Data Type
depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	Feedback Category
parent	Feedback Category

Feedback Comment

Field Name	Data Type
content	string
date_created	datetime
email	string
feedback_id	number
id	number
ip_address	string
is_reviewed	boolean

name	string
person_id	number
status	string
validating	string
visitor_id	number
website	string
feedback	Feedback
person	Person
visitor	Visitor

Feedback Revision

Field Name	Data Type
content	string
date_created	datetime
feedback_id	number
id	number
person_id	number
status	string
title	string
feedback	Feedback
person	Person

Feedback Status Category

Field Name	Data Type
display_order	number
id	number
status_type	string
title	string

Label Article

Field Name	Data Type
article_id	number
label	string
article	Article

Label Blob

Field Name	Data Type
blob_id	number
label	string
blob	Blob

Label Chat Conversation

Field Name	Data Type
chat_id	number
label	string
chat	Chat Conversation

Label Download

Field Name	Data Type
download_id	number
label	string
download	Download

Label Feedback

Field Name	Data Type
feedback_id	number
label	string
feedback	Feedback

Label News

Field Name	Data Type
label	string
news_id	number
news	News

Label Organization

Field Name	Data Type
label	string
organization_id	number
organization	Organization

Label Person

Field Name	Data Type
label	string
person_id	number
person	Person

Label Task

Field Name	Data Type
label	string
task_id	number
task	Task

Label Ticket

Field Name	Data Type
label	string
ticket_id	number
ticket	Ticket

Language

Field Name	Data Type
base_filepath	string
flag_image	string
has_admin	boolean
has_agent	boolean
has_user	boolean
id	number
is_rtl	boolean
lang_code	string
locale	string
sys_name	string
title	string

News

Field Name	Data Type
category_id	number
content	string
date_created	datetime
date_published	datetime
hidden_status	string
id	number
language_id	number
num_comments	number
num_ratings	number
person_id	number
slug	string
status	string
title	string
total_rating	number
view_count	number

category	News Category
labels	Label News
language	Language
person	Person
revisions	News Revision

News Category

Field Name	Data Type
depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	News Category
parent	News Category

News Comment

Field Name	Data Type
content	string
date_created	datetime
email	string
id	number
ip_address	string
is_reviewed	boolean
name	string
news_id	number
person_id	number
status	string
validating	string

visitor_id	number
website	string
news	News
person	Person
visitor	Visitor

News Revision

Field Name	Data Type
content	string
date_created	datetime
id	number
news_id	number
person_id	number
status	string
title	string
news	News
person	Person

Organization

Field Name	Data Type
date_created	datetime
id	number
importance	number
name	string
picture_blob_id	number
summary	string
contact_data	Organization Contact Data
custom_data	Custom Data Organization
custom_data[#]	Custom Data Organization (Gets data for the organization field with ID #)
email_domains	Organization Email Domain

labels	Label Organization
picture_blob	Blob

Organization Contact Data

Field Name	Data Type
comment	string
contact_type	string
field_1	string
field_2	string
field_3	string
field_4	string
field_5	string
field_6	string
field_7	string
field_8	string
field_9	string
field_10	string
id	number
organization_id	number
organization	Organization

Organization Email Domain

Field Name	Data Type
domain	string
organization_id	number
organization	Organization

Person

Field Name	Data Type
can_admin	boolean

can_agent	boolean
can_billing	boolean
can_reports	boolean
creation_system	string
date_created	datetime
date_last_login	datetime
date_picture_check	datetime
disable_autoresponses	boolean
first_name	string
gravatar_url	string
id	number
importance	number
is_agent	boolean
is_agent_confirmed	boolean
is_confirmed	boolean
is_contact	boolean
is_deleted	boolean
is_disabled	boolean
is_user	boolean
is_vacation_mode	boolean
language_id	number
last_name	string
name	string
organization_id	number
organization_manager	boolean
organization_position	string
override_display_name	string
picture_blob_id	number
primary_email_id	number
summary	string

timezone	string
title_prefix	string
was_agent	boolean
contact_data	Person Contact Data
custom_data	Custom Data Person
custom_data[#]	Custom Data Person (Gets data for the person field with ID #)
emails	Person Email
labels	Label Person
language	Language
organization	Organization
picture_blob	Blob
preferences	Person Pref
primary_email	Person Email
usersource_assoc	Person Usersource Assoc

Person Contact Data

Field Name	Data Type
comment	string
contact_type	string
field_1	string
field_2	string
field_3	string
field_4	string
field_5	string
field_6	string
field_7	string
field_8	string
field_9	string
field_10	string
id	number

person_id	number
person	Person

Person Email

Field Name	Data Type
comment	string
date_created	datetime
date_validated	datetime
email	string
email_domain	string
id	number
is_validated	boolean
person_id	number
person	Person

Person Email Validating

Field Name	Data Type
auth	string
date_created	datetime
email	string
id	number
person_id	number
validating_content	string
person	Person

Person Pref

Field Name	Data Type
date_expire	datetime
name	string
person_id	number

value_array	string
value_str	string
person	Person

Person Usersource Assoc

Field Name	Data Type
data	string
date_created	datetime
id	number
identity	string
identity_friendly	string
person_id	number
usersource_id	number
person	Person
usersource	Usersource

Plugin

Field Name	Data Type
date_created	datetime
description	string
enabled	boolean
id	string
package_class	string
package_class_file	string
resources_path	string
title	string
version	string

Product

Field Name	Data Type

depth	number
display_order	number
id	number
parent_id	number
root	number
title	string
children	Product
parent	Product

Session

Field Name	Data Type
active_status	string
auth	string
data	string
date_created	datetime
date_last	datetime
id	number
is_bot	boolean
is_chat_available	boolean
is_helpdesk	boolean
is_person	boolean
page_count	number
person_id	number
visitor_id	number
person	Person
visitor	Visitor

SLA

Field Name	Data Type
active_time	string

allow_agent_manual	boolean
apply_all	boolean
apply_priority_id	number
apply_trigger_id	number
fail_trigger_id	number
id	number
sla_type	string
title	string
warning_trigger_id	number
work_days	string
work_end	number
work_holidays	string
work_start	number
work_timezone	string
apply_priority	Ticket Priority
apply_trigger	Ticket Trigger
fail_trigger	Ticket Trigger
ticket_slas	Ticket Sla
ticket_slas[#]	Ticket Sla (Gets ticket SLA data for the SLA with ID #)
warning_trigger	Ticket Trigger

Task

Field Name	Data Type
assigned_agent_id	number
assigned_agent_team_id	number
date_completed	datetime
date_created	datetime
date_due	date
id	number
is_completed	boolean

person_id	number
title	string
visibility	number
assigned_agent	Person
assigned_agent_team	Agent Team
comments	Task Comment
labels	Label Task
person	Person

Task Comment

Field Name	Data Type
content	string
date_created	datetime
id	number
person_id	number
task_id	number
person	Person
task	Task

Ticket

Field Name	Data Type
agent_id	number
agent_team_id	number
auth	string
category_id	number
creation_system	string
creation_system_option	string
date_agent_waiting	datetime
date_closed	datetime
date_created	datetime

date_feedback_rating	datetime
date_first_agent_assign	datetime
date_first_agent_reply	datetime
date_last_agent_reply	datetime
date_last_user_reply	datetime
date_locked	datetime
date_resolved	datetime
date_status	datetime
date_user_waiting	datetime
department_id	number
email_gateway_address_id	number
email_gateway_id	number
feedback_rating	number
has_attachments	boolean
hidden_status	string
id	number
is_hold	boolean
language_id	number
linked_chat_id	number
locked_by_agent	number
notify_email	string
notify_email_agent	string
notify_email_name	string
notify_email_name_agent	string
organization_id	number
person_email_id	number
person_email_validating_id	number
person_id	number
priority_id	number
product_id	number

ref	string
sent_to_address	string
status	string
subject	string
ticket_hash	string
total_to_first_reply	number
total_user_waiting	number
urgency	number
validating	string
waiting_times	string
workflow_id	number
worst_sla_status	string
access_codes	Ticket Access Code
agent	Person
agent_team	Agent Team
attachments	Ticket Attachment
category	Ticket Category
charges	Ticket Charge
custom_data	Custom Data Ticket
custom_data[#]	Custom Data Ticket (Gets data for the ticket field with ID #)
department	Department
email_gateway	Email Gateway
email_gateway_address	Email Gateway Address
labels	Label Ticket
language	Language
linked_chat	Chat Conversation
locked_by_agent	Person
messages	Ticket Message
organization	Organization
participants	Ticket Participant

person	Person
person_email	Person Email
person_email_validating	Person Email Validating
priority	Ticket Priority
product	Product
ticket_slas	Ticket Sla
ticket_slas[#]	Ticket Sla (Gets ticket SLA data for the SLA with ID #)
workflow	Ticket Workflow

Ticket Access Code

Field Name	Data Type
auth	string
id	number
person_id	number
ticket_id	number
person	Person
ticket	Ticket

Ticket Attachment

Field Name	Data Type
blob_id	number
id	number
is_agent_note	boolean
is_inline	boolean
message_id	number
person_id	number
ticket_id	number
blob	Blob
message	Ticket Message
person	Person

ticket	Ticket

Ticket Category

Field Name	Data Type
display_order	number
id	number
parent_id	number
title	string
children	Ticket Category
parent	Ticket Category

Ticket Charge

Field Name	Data Type
agent_id	number
amount	number
charge_time	number
comment	string
date_created	datetime
id	number
organization_id	number
person_id	number
ticket_id	number
agent	Person
organization	Organization
person	Person
ticket	Ticket

Ticket Feedback

Field Name	Data Type
date_created	datetime

id	number
message	string
message_id	number
person_id	number
rating	number
ticket_id	number
person	Person
ticket	Ticket
ticket_message	Ticket Message

Ticket Log

Field Name	Data Type
action_type	string
date_created	datetime
details	string
id	number
id_after*	number
id_before	number
id_object	number
person_id	number
sla_id	number
sla_status	string
ticket_id	number
trigger_id	number
person	Person
sla	Sla
ticket	Ticket

- id_after means the ticket ID after it has changed; e.g. if this log represents a change from ID X to ID Y, id_after = Y.

Ticket Message

Field Name	Data Type
creation_system	string
date_created	datetime
email	string
email_source_id	number
id	number
ip_address	string
is_agent_note	boolean
message	string
message_full	string
message_hash	string
message_raw	string
person_id	number
show_full_hint	boolean
ticket_id	number
visitor_id	number
attachments	Ticket Attachment
email_source	Email Source
person	Person
ticket	Ticket
visitor	Visitor

Ticket Participant

Field Name	Data Type
access_code_id	number
default_on	boolean
id	number
person_email_id	number
person_id	number
ticket_id	number

access_code	Ticket Access Code
person	Person
person_email	Person Email
ticket	Ticket

Ticket Priority

Field Name	Data Type
id	number
priority	number
title	string

Ticket SLA

Field Name	Data Type
fail_date	datetime
id	number
is_completed	boolean
sla_id	number
sla_status	string
ticket_id	number
warn_date	datetime
sla	Sla
ticket	Ticket

Ticket Trigger

Field Name	Data Type
actions	string
date_created	datetime
event_trigger	string
event_trigger_options	string
id	number

is_enabled	boolean
run_order	number
sys_name	string
terms	string
terms_any	string
title	string

Ticket Workflow

Field Name	Data Type
display_order	number
id	number
title	string

Usersource

Field Name	Data Type
display_order	number
id	number
is_enabled	boolean
lost_password_url	string
options	string
source_type	string
title	string
usersource_plugin_id	number
usersource_plugin	Usersource Plugin

Usersource Plugin

Field Name	Data Type
adapter_class	string
form_model_class	string
form_template	string

form_type_class	string
id	number
plugin_id	number
title	string
unique_key	string
plugin	Plugin

Visitor

Field Name	Data Type
auth	string
date_created	datetime
date_last	datetime
email	string
id	number
ip_address	string
landing_page	string
last_page	string
name	string
person_id	number
ref_page	string
session_landing_page	string
user_agent	string
person	Person